

Physical Security — Guided Notes ANSWER KEY

Unit 4.4.1 | CompTIA Network+ N10-009 Obj. 4.4 | N10-008 4.4

For instructor use / distribute after students complete the notes.

Question 1

Physical access to a switch provides an attacker with _____ access, bypassing all software security controls. Explain how an attacker with physical access to a Cisco switch can recover the password without knowing it.

Answer: Console (direct hardware); password recovery: boot into ROMMON mode, change the config register to bypass startup config, reload and set a new password — documented in Cisco's own support guides.

Question 2

An access control vestibule (mantrap) prevents _____ by requiring each person to badge through a first door that must _____ before the inner door can open, limiting entry to one person at a time.

Answer: Tailgating (piggybacking); close/lock.

Question 3

Badge systems log every _____ with a timestamp, creating an audit trail. When an employee is terminated, their badge should be disabled _____. What does tailgating defeat that a badge system cannot prevent on its own?

Answer: Entry (access event); immediately. Tailgating defeats the badge requirement — an authorized person opening the door provides physical access to someone who hasn't badged in.

Question 4

Server rooms require environmental monitoring for _____ (should stay between 64-75°F), _____ (40-60% relative humidity), and _____ detection. Why is humidity too low (below 40%) dangerous for network equipment?

Answer: Temperature; humidity; water (leak). Low humidity increases static electricity discharge risk, which can permanently damage sensitive electronic components.

Question 5

Fire suppression in a server room must use _____ agents rather than water sprinklers. Name one clean agent system used in data centers and explain why water suppression is avoided.

Answer: Clean (gaseous); examples: FM-200, Novec 1230, inert gas (IG-541). Water destroys electronic equipment and creates electrical hazards — clean agents extinguish fire by removing heat or displacing oxygen without leaving residue.

Question 6

Drive disposal must match the sensitivity of the data. For standard data, _____ (multiple-pass overwrite) meets most requirements. For SSDs, overwriting is unreliable because of _____, so the

recommended approach is _____ or physical destruction.

Answer: Drive wiping (e.g., DBAN, DoD 5220.22-M); wear leveling (the controller may not overwrite every physical cell); manufacturer's secure erase command.

Question 7

Degaussing uses a strong _____ field to destroy data on magnetic media. It renders the media _____ (cannot be reused). Degaussing does NOT work on _____ because they store data electronically, not magnetically.

Answer: Magnetic; permanently unusable; SSDs (solid-state drives).

Question 8 — Real World Application

REAL WORLD: A contractor arrives to service the HVAC system in the server room. The front desk gives them a visitor badge and points them toward the server room. They enter unsupervised for 45 minutes. Identify two policy failures and describe the correct procedure.

Answer: Failure 1: No escort — visitors in sensitive areas must be accompanied by authorized IT staff at all times. Failure 2: No access logging/sign-in for the server room entry. Correct procedure: IT staff meets the contractor at the front desk, escorts them throughout the visit, and logs entry/exit time and purpose in the access record.